

UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy
DEGREE OF M.Sci. WITH HONOURS
FOURTH-YEAR FINAL EXAMINATION

03 17307

LM Superconductivity

SEMESTER 1 EXAMINATIONS 2022/23

Time Allowed: 1 hour 30 minutes

Answer **two** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets [].

All symbols have their usual meanings.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and a table of physical constants and units that may be required will be found at the end of this question paper.

Answer **two** questions. If you answer more than two questions, credit will only be given for the best two answers.

1. Consider a thin film of superconductor in a parallel magnetic field, where the thickness is smaller than the coherence length, $d < \xi(T)$. The film is defined by the plane boundaries $z = \pm d/2$, and the applied field $\mathbf{B} = B_0 \mathbf{j}$ is along the y -direction, where $\mathbf{j}$ is the unit vector along the y -direction. Since $d < \xi$, the order parameter amplitude $|\psi| = g\psi_0$ is constant across the sample, where ψ_0 is the bulk order parameter amplitude and g is the dimensionless order parameter.

- (a) Starting from London's equation,

$$\mathbf{J}_s = \frac{n_s^* e^*}{m^*} (\hbar \nabla \phi - e^* \mathbf{A}),$$

show that the magnetic field distribution in the film has the form

$$B_y(z) = B_0 \frac{\cosh(zg/\lambda)}{\cosh(dg/2\lambda)},$$

where λ is the bulk penetration depth.

[8]

- (b) Show that the supercurrent density distribution in the film has the form

$$J_x(z) = -\frac{B_0 g}{\mu_0 \lambda} \frac{\sinh(zg/\lambda)}{\cosh(dg/2\lambda)}.$$

[6]

- (c) Show that the mean square value of the supercurrent velocity in the film is given by

$$\overline{v_s^2} = \frac{1}{d} \int_{-d/2}^{d/2} v(z)^2 dz = \frac{1}{2} \left(\frac{B_0 e^* \lambda}{m^* g} \right)^2 \frac{\left[\frac{\lambda}{dg} \sinh\left(\frac{dg}{\lambda}\right) - 1 \right]}{\cosh^2\left(\frac{dg}{2\lambda}\right)}.$$

[6]

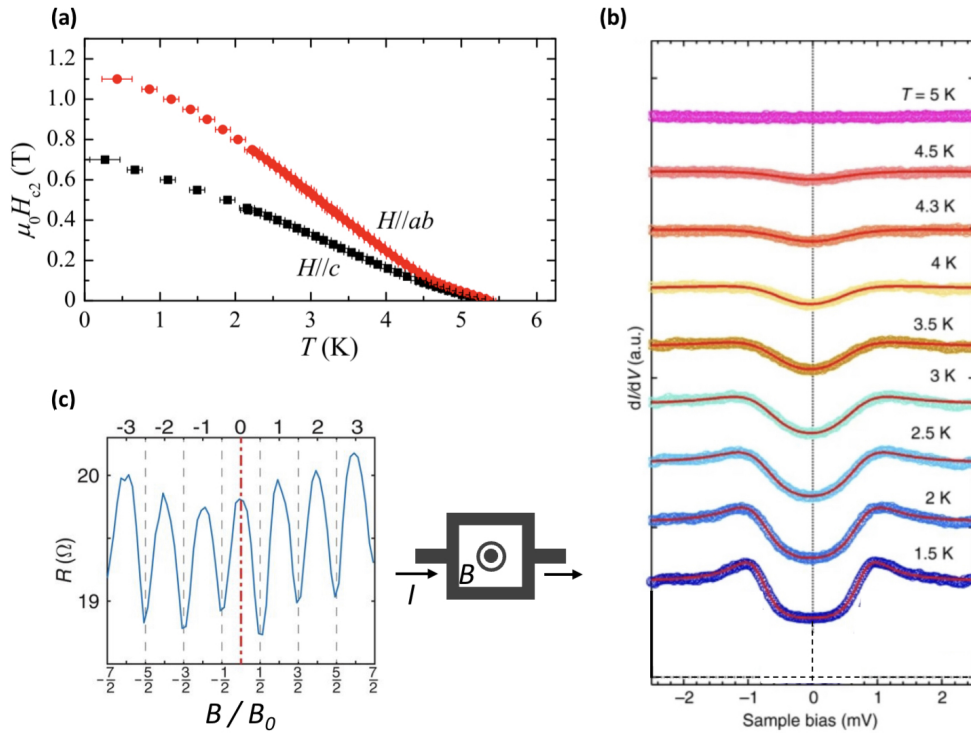
- (d) By minimising the Ginzburg-Landau free energy with respect to g^2 , show that the parameter g satisfies the equation

$$g^2 = 1 - \frac{m^* \overline{v_s^2}}{2|a|},$$

where a is a Ginzburg-Landau parameter. Briefly explain the meaning of the equation.

[10]

2. Suppose you are given a type-II superconductor and measure its various properties to obtain the results as shown in Figures (a)-(c).



- (a) Figure (a) plots the second critical field strength, H_{c2} , as a function of temperature for the field applied along the crystallographic c axis and along the ab plane of the superconductor. Estimate the coherence length, ξ , at zero temperature within the Ginzburg-Landau framework, for both $H \parallel c$ and $H \parallel ab$, and briefly explain how to obtain H_{c2} values experimentally. [6]
- (b) Figure (b) plots the differential conductance, dI/dV , as a function of the bias voltage for different temperatures. The data are shifted vertically for better visibility. Estimate the superconducting gap parameter, Δ , at zero temperature, and briefly discuss your method of estimation. Use the available data to discuss whether the result agrees with a BCS prediction. [12]
- (c) Suppose you make a rectangular ring from this superconductor and flow currents (I) while varying an applied magnetic field (B) that threads the ring, as shown on the right in Figure (c). The resistance (R) thus obtained is shown on the left in Figure (c), where $B_0 = 3$ mT. Draw the expected T_c profile as a function of B/B_0 . Estimate the dimension of the rectangular

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ring. Point out and discuss any anomalous feature in the shown resistance data.

[12]

3. Suppose you measured the magnetic susceptibility, χ_v , and molar specific heat, C , of a superconductor as a function of temperature, T , and obtained the data as shown in Figures (a)-(c) below.

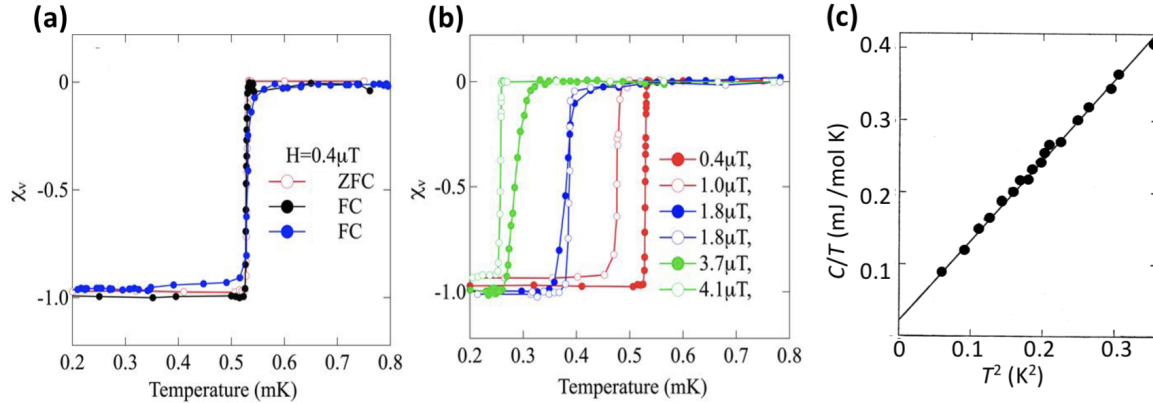


Figure (a) plots $\chi_v(T)$ in an externally applied field of $0.4 \mu\text{T}$ under zero-field-cooled (ZFC; red circles) and field-cooled (FC; black and blue circles) conditions. For ZFC, the superconductor is first cooled down in zero field across the superconducting transition at T_c to the lowest measured temperature, where the field is applied and $\chi_v(T)$ is subsequently measured while warming up. For FC, the field is first applied to the superconductor above T_c , cooled down to the lowest measured temperature, then $\chi_v(T)$ is measured while warming up. Figure (b) shows $\chi_v(T)$ at different applied field strengths. Figure (c) shows the data for $C(T)$ measurements at zero field.

- (a) From Figure (a), discuss whether this superconductor is type I or type II with brief reasoning. [5]

- (b) Using the data from Figure (b), draw a field-temperature phase diagram and estimate the critical field strength at zero temperature. [5]

- (c) Suppose that a separate experiment on this superconductor finds the carrier density $n \simeq 3 \times 10^{17} \text{ cm}^{-3}$ and the mean free path $l \sim 300 \mu\text{m}$ in the normal state. Discuss whether this superconductor belongs to a clean or dirty limit or otherwise. (Hint: You need to estimate another relevant quantity.) [10]

- (d) From Figure (c), estimate the Debye temperature, Θ_D . Using experimental data available from the Figures and relevant formula within the BCS framework,

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discuss whether this superconductor is in the weak or strong coupling limit or otherwise.

[10]

Formula Sheet

The molar specific heat of metals at low temperature is described by

$$C = \gamma T + \left(\frac{12}{5}\right) \pi^4 R \left(\frac{T}{\Theta_D}\right)^3,$$

where γ is the Sommerfeld constant, R the gas constant, and Θ_D the Debye temperature.

The Fermi velocity in metals is given by

$$v_F = \left(\frac{\hbar}{m_e}\right) (3\pi^2 n)^{1/3},$$

where m_e is the mass of an electron and n the density of charge carriers.

Some expressions from the Ginzburg-Landau (GL) theory:

$$\lambda = \sqrt{m^* b / 4\mu_0 e^2 |a|},$$

$$\xi = \sqrt{\hbar^2 / 2m^* |a|},$$

$$B_{c2} = \frac{\Phi_0}{2\pi\xi^2} = \sqrt{2}\kappa B_c,$$

where a and b are the GL parameters, Φ is the flux quantum and $\kappa \equiv \lambda/\xi$.

The GL free energy density difference is expressed as

$$f_s - f_n = a|\psi|^2 + \frac{1}{2}b|\psi|^4 + \frac{1}{2m^*} |(-i\hbar - e^* \mathbf{A}) \psi|^2,$$

where f_s and f_n are the free energy density in the superconducting and normal states, respectively. Also note that $n^* = |\psi|^2$.

Some predictions from the BCS theory:

$$\xi_0 = \frac{\hbar v_F}{3.52 k_B T_c},$$

$$\Delta_0 = 1.76 k_B T_c,$$

$$k_B T_c = 1.14 k_B \Theta_D \exp[-1/D(E_F)V],$$

where $D(E_F)$ is the density of states at the Fermi level and V the effective electron-electron attraction via phonons.

Physical Constants and Units

Acceleration due to gravity	g	9.81 m s^{-2}
Gravitational constant	G	$6.674 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Ice point	T_{ice}	273.15 K
Avogadro constant	N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
[N.B. 1 mole $\equiv$ 1 gram-molecule]		
Gas constant	R	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23} \text{ J K}^{-1} \equiv 8.62 \times 10^{-5} \text{ eV K}^{-1}$
Stefan constant	σ	$5.670 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Rydberg constant	R_∞	$1.097 \times 10^7 \text{ m}^{-1}$
	$R_\infty hc$	13.606 eV
Planck constant	h	$6.626 \times 10^{-34} \text{ J s} \equiv 4.136 \times 10^{-15} \text{ eV s}$
	$h/2\pi$	$\hbar$ $1.055 \times 10^{-34} \text{ J s} \equiv 6.582 \times 10^{-16} \text{ eV s}$
Speed of light <i>in vacuo</i>	c	$2.998 \times 10^8 \text{ m s}^{-1}$
	$\hbar c$	197.3 MeV fm
Charge of proton	e	$1.602 \times 10^{-19} \text{ C}$
Mass of electron	m_e	$9.109 \times 10^{-31} \text{ kg}$
Rest energy of electron		0.511 MeV
Mass of proton	m_p	$1.673 \times 10^{-27} \text{ kg}$
Rest energy of proton		938.3 MeV
One atomic mass unit	u	$1.66 \times 10^{-27} \text{ kg}$
Atomic mass unit energy equivalent		931.5 MeV
Electric constant	ϵ_0	$8.854 \times 10^{-12} \text{ F m}^{-1}$
Magnetic constant	μ_0	$4\pi \times 10^{-7} \text{ H m}^{-1}$
Bohr magneton	μ_B	$9.274 \times 10^{-24} \text{ A m}^2 (\text{J T}^{-1})$
Nuclear magneton	μ_N	$5.051 \times 10^{-27} \text{ A m}^2 (\text{J T}^{-1})$
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	$7.297 \times 10^{-3} = 1/137.0$
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12} \text{ m}$
Bohr radius	a_0	$5.2918 \times 10^{-11} \text{ m}$
angstrom	$\text{\AA}$	10^{-10} m
barn	b	10^{-28} m^2
torr (mm Hg at 0 °C)	torr	$133.32 \text{ Pa (N m}^{-2})$

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Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches must be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.